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WIRE HARNESS FIXING DEVICE AND WIRE HARNESS

Abstract

A wire harness fixing tool (**13**) according to one embodiment of the present disclosure comprises a first holding member (**20**) for holding a first electrical wire member (**11**) and a second holding member (**30**) for holding a second electrical wire member (**12**). The second holding member (**30**) is configured so as to be detachably attachable to the first holding member (**20**).

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a wire harness fixing device and a wire harness.

BACKGROUND

[0002] For example, the wire harness fixing device described in Patent Document 1 includes a fixed part that is fixed to a vehicle body panel or the like, and a holding part that holds electric wire members. The electric wire members are arranged by this fixing device along the vehicle body panel.

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: JP H07-322449 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0004] The configuration of electric wire members required for a vehicle, such as the type and number, varies depending on differences in vehicle equipment according to the grade of the vehicle, the presence or absence of selected optional items, and the like. For this reason, the present inventor has studied how the configuration of the electric wire members can be easily changed.

[0005] An object of the present disclosure is to provide a wire harness fixing device and a wire harness in which the configuration of electric wire members can be easily changed.

Means to Solve the Problem

[0006] A wire harness fixing device according to an aspect of the present disclosure includes a first holding member that holds a first electric wire member and a second holding member that holds a second electric wire member, and the second holding member is attachable to and detachable from the first holding member.

[0007] A wire harness according to an aspect of the present disclosure includes the above-described wire harness fixing device, a first electric wire member that is held by the first holding member of the wire harness fixing device, and a second electric wire member that is held by the second holding member of the wire harness fixing device.

Effect of the Invention

[0008] According to an aspect of the present disclosure, it is possible to provide a wire harness fixing device and a wire harness in which the configuration of electric wire members can be easily changed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a wire harness in an embodiment.

[0010] FIG. 2 is a schematic diagram of the wire harness in the same embodiment.

[0011] FIG. 3 is an exploded perspective view of a wire harness fixing device in the same embodiment.

[0012] FIG. 4 is a schematic perspective view of a second holding member in the same embodiment.

[0013] FIG. 5 is a schematic perspective view of the second holding member in the same embodiment.

[0014] FIG. 6 is a schematic diagram of the wire harness in the same embodiment.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Description of Embodiments of Present Disclosure

[0015] First, embodiments of the present disclosure will be listed and described.

[0016] A wire harness fixing device of the present disclosure includes a first holding member that holds a first electric wire member, and a second holding member that holds a second electric wire member, and the second holding member is attachable to and detachable from the first holding member.

[0017] With this configuration, after removing the second holding member and the second electric wire member from the first holding member that is fixed to a target object, it is possible to fix the second holding member holding a new second electric wire member to the first holding member, for example. In addition, if the first electric wire member is a standard-grade electric wire member and the second electric wire member is an electric wire member corresponding to an optional function, for example, it is possible to correspond to whether or not the option has been selected by attaching and detaching the second holding member. In this manner, the above-described wire harness fixing device makes it possible to easily change the configuration of the electric wire members.

[0018] [2] In the above [1], the first holding member may have a fixed part that is fixed to an attachment target and an electric wire holding part that holds the first electric wire member.

[0019] This configuration makes it possible to attach and detach the second holding member to and from the first holding member that is fixed to the attachment target.

[0020] [3] In the above [2], the electric wire holding part may be inseparable from the fixed part.

[0021] This configuration makes it possible to attach and detach the second holding member to and from the first holding member in which the electric wire holding part is inseparable from the fixed part.

[0022] [4] In any of the above [1] to [3], one of the first holding member and the second holding member may have an engaging part, the other of the first holding member and the second holding member may have an engaged part with which the engaging part engages, and the second holding member may be fixed to the first holding member by engagement between the engaging part and the engaged part.

[0023] This configuration makes the second holding member attachable to and detachable from the first holding member by the engagement between the engaging part and the engaged part.

[0024] [5] In the above [4], the engaged part may have an insertion hole, and the engaging part may have a pillar portion that is inserted into the insertion hole and a protruding portion that protrudes from the pillar portion and is locked to the engaged part in a counter-insertion direction.

[0025] This configuration makes it possible to make the engaging part attachable to and detachable from the engaged part.

[0026] In the above [5], a plurality of the engaging parts and a plurality of the insertion holes may be provided.

[0027] With this configuration, the second holding member can be firmly fixed to the first holding member by the plurality of sets of engaging parts and engaged parts.

[0028] [7] In any of the above [4] to [6], the engaged part may have an insertion hole into which the engaging part is inserted, and the engaged part may be provided in the first holding member.

[0029] This configuration makes it possible to, when the first holding member is used without the second holding member attached, prevent the engaged part from interfering with surrounding components.

[0030] [8] In any of the above [5] to [7], the protruding portion may have a plate-like shape that is elastically deformable in a direction along the insertion direction with respect to the insertion hole.

[0031] With this configuration, the engaging part including the protruding portion can be suitably attached to and detached from the engaged part.

[0032] In any of the above [5] to [7], the protruding portion may form a spherical or ellipsoidal shape.

[0033] With this configuration, the engaging part including the protruding portion can be suitably

attached to and detached from the engaged part.

[0034] In any of the above [5] to [9], the protruding portion may be integrally formed with the pillar portion.

[0035] This configuration makes it possible to improve the locking strength of the protruding portion to the engaged part.

[0036] In any of the above [5] to [9], the protruding portion may be configured separately from the pillar portion.

[0037] This configuration makes it possible to improve the formability of the engaging part including the protruding portion.

[0038] In any of the above [4] to [11], the first holding member may have a fixed part that is fixed to an attachment target and an electric wire holding part that holds the first electric wire member, the electric wire holding part may have a band portion that surrounds an outer periphery of the first electric wire member and a lock portion by which the band portion is locked, and in the first holding member, the engaging part or the engaged part may be provided between the lock portion and the fixed part.

[0039] This configuration makes it possible to preferably configure the engaging part or the engaged part in the first holding member.

[0040] In the above [1], the first holding member may be a binding member that binds an electric wire bundle of the first electric wire member.

[0041] This configuration makes it possible to attach and detach the second holding member to and from the binding member.

[0042] In the above [13], the binding member may be an adhesive tape that is wound around an outer periphery of the electric wire bundle.

[0043] This configuration makes it possible to attach and detach the second holding member to and from the adhesive tape.

[0044] A wire harness according to the present disclosure includes the wire harness fixing device according to any one of the above [1] to [14], a first electric wire member that is held by the first holding member of the wire harness fixing device, and a second electric wire member that is held by the second holding member of the wire harness fixing device.

[0045] This configuration makes it possible to obtain the same advantageous effect as that of the wire harness fixing device in the above [1].

Details of Embodiments of Present Disclosure

[0046] Specific examples of a wire harness fixing device and a wire harness of the present disclosure will be described below with reference to the drawings. In each drawing, for convenience of description, a part of the configuration may be exaggerated or simplified. The dimensional ratio of each part may differ among the drawings. The terms “parallel” and “perpendicular” herein do not only mean strictly parallel or perpendicular, but also include roughly parallel or perpendicular within the range in which the operations and effects of the present embodiment can be achieved.

[0047] As shown in FIG. 1, a wire harness **10** of the present embodiment includes a first electric wire member **11**, a second electric wire member **12**, and a wire harness fixing device **13**. The wire harness fixing device **13** has a first holding member **20** and a second holding member **30**. The first electric wire member **11** is held by the first holding member **20**. The second electric wire member **12** is held by the second holding member **30**. The second holding member **30** is attachable to and detachable from the first holding member **20**.

[0048] Each of the first electric wire member **11** and the second electric wire member **12** may be a bundle of a plurality of electric wires or may include an electric wire bundle, for example. Each of the first electric wire member **11** and the second electric wire member **12** may include a cylindrical exterior member that covers the outer periphery of the electric wire bundle.

Configuration of First Holding Member **20**

[0049] As shown in FIG. 2, the first holding member 20 has a fixed part 21 that is fixed to a vehicle body panel P as an attachment target, a first electric wire holding part 22 that holds the first electric wire member 11, and an engaged part 23. The first holding member 20 is formed from a synthetic resin, for example. The first holding member 20 is an injection-molded article in which the fixed part 21, the first electric wire holding part 22, and the engaged part 23 are integrally formed, for example. That is, the first electric wire holding part 22 is unattachable to and undetachable from the fixed part 21.

[0050] The fixed part 21 is fixed by being inserted into an attachment hole Pa that is formed through the vehicle body panel P, for example. The fixed part 21 includes a pair of projecting pieces 21a and a flange portion 21b, for example. The flange portion 21b is a suction cup-shaped portion, for example. The fixed part 21 sandwiches the vehicle body panel P between the projecting pieces 21a and the flange portion 21b. In this manner, the fixed part 21 is fixed to the vehicle body panel P.

[0051] As shown in FIGS. 1 and 2, the first electric wire holding part 22 has a band portion 24 that surrounds the outer periphery of the first electric wire member 11, and a lock portion 25 by which the band portion 24 is locked. The lock portion 25 is formed at one end in the length direction of the band portion 24. The band portion 24 is wrapped around the first electric wire member 11, and a leading end of the band portion 24 is inserted into and locked to the lock portion 25. In this manner, the first electric wire member 11 is held by the band portion 24 and the lock portion 25.

(Configuration of Engaged Part 23)

[0052] As shown in FIGS. 2 and 3, the engaged part 23 is provided between the lock portion 25 and the fixed part 21. The engaged part 23 is connected to the lock portion 25 and the flange portion 21b, for example. The engaged part 23 includes a wall portion 27 having an insertion hole 26. For example, two insertion holes 26 are provided in the wall portion 27. In the drawings, the X axis is an axis along the length direction of the first electric wire member 11. The Y axis is an axis perpendicular to the X axis and parallel to the surface of the vehicle body panel P. The Z axis is an axis along the height direction of the first holding member 20 and perpendicular to both the X axis and the Y axis. For example, the wall portion 27 is a wall perpendicular to the Y axis, and the insertion holes 26 pass through the wall portion 27 along the Y axis.

Configuration of Second Holding Member 30

[0053] The second holding member 30 includes a second electric wire holding part 31 that holds the second electric wire member 12, and an engaging part 32 that engages with the engaged part 23. The second holding member 30 is formed from a synthetic resin, for example. The second holding member 30 is an injection-molded article in which the second electric wire holding part 31 and the engaging part 32 are integrally formed, for example. For example, two engaging parts 32 are provided. The engaging parts 32 are attachable to and detachable from the engaged part 23 of the first holding member 20.

[0054] As shown in FIGS. 1 and 2, the second electric wire holding part 31 has a band portion 33 that surrounds the outer periphery of the second electric wire member 12, and a lock portion 34 by which the band portion 33 is locked. The lock portion 34 is formed at one end in the length direction of the band portion 33. The band portion 33 is wrapped around the second electric wire member 12, and a leading end of the band portion 33 is inserted into and locked to the lock portion 34. In this manner, the second electric wire member 12 is held by the band portion 33 and the lock portion 34.

Configuration of Engaging Part 32

[0055] As shown in FIGS. 2 and 3, the engaging part 32 has a pillar portion 35 that is inserted into the insertion hole 26, and a protruding portion 36 that protrudes from the pillar portion 35. The pillar portion 35 extends along the Y axis from a base portion 37 that is integrally formed with the lock portion 34, for example. The protruding portion 36 protrudes in the radial direction of the pillar portion 35 in a direction perpendicular to the extending direction of the pillar portion 35, for

example. For example, two protruding portions **36** are provided. The protruding portions **36** are integrally formed with the pillar portion **35**, for example.

[0056] The protruding portions **36** are in the form of a plate that is elastically deformable in a direction along an insertion direction D with respect to the insertion holes **26**, for example. The insertion direction D with respect to the insertion holes **26** is a direction along the Y axis. The protruding portions **36** are in the form of a plate that is perpendicular to or intersects with the insertion direction D so as to bend in a direction along the insertion direction D. The plate thickness of the protruding portions **36** in the insertion direction D is made small so as to bend suitably in the direction along the insertion direction D.

[0057] The engaging parts **32** are inserted into the insertion holes **26**. When the engaging parts **32** are respectively inserted into the insertion holes **26**, the protruding portions **36** pass through the insertion holes **26** while bending. After passing through the insertion holes **26**, the protruding portions **36** elastically return to their original shape. Accordingly, the protruding portions **36** are locked to the wall portion **27** of the engaged part **23** in a counter-insertion direction. The counter-insertion direction is a direction opposite to the insertion direction D and along the Y axis. The second holding member **30** is fixed to the first holding member **20** by the engagement between the engaging parts **32** and the wall portion **27** of the engaged part **23**.

[0058] The operations of the present embodiment will be described.

[0059] The second holding member **30** that holds the second electric wire member **12** is attachable to and detachable from the first holding member **20** that holds the first electric wire member **11**. Therefore, it is possible to remove the second electric wire member **12** from the first holding member **20** without removing the first holding member **20** from the vehicle body panel P. In addition, since the second holding member **30** is attachable to the first holding member **20** that is fixed to the vehicle body panel P, the second electric wire member **12** can be easily replaced.

[0060] The advantageous effects of the present embodiment will be described.

[0061] (1) The second holding member **30** that holds the second electric wire member **12** is attachable to and detachable from the first holding member **20** that holds the first electric wire member **11**. With this configuration, after removing the second holding member **30** and the second electric wire member **12** from the first holding member **20** that is fixed to the vehicle body panel P, it is possible to newly fix the second holding member **30** holding another second electric wire member **12** with different specifications to the first holding member **20**, for example. In addition, if the first electric wire member **11** is a standard grade electric wire member and the second electric wire member **12** is an electric wire member corresponding to an optional function, for example, it is possible to correspond to whether or not the option has been selected by attaching and detaching the second holding member **30**. It is also possible to reuse the second holding member **30** and the second electric wire member **12** removed from the first holding member **20**, in another vehicle. As above, the wire harness fixing device **13** makes it possible to easily change the configuration of the electric wire members mounted on the vehicle.

[0062] (2) The first holding member **20** includes the fixed part **21** that is fixed to the vehicle body panel P as the attachment target, and the first electric wire holding part **22** that holds the first electric wire member **11**. This configuration makes it possible to attach and detach the second holding member **30** to and from the first holding member **20** that is fixed to the vehicle body panel P.

[0063] (3) The first electric wire holding part **22** is inseparable from the fixed part **21**. This configuration makes it possible to attach and detach the second holding member **30** to and from the first holding member **20** in which the first electric wire holding part **22** is inseparable from the fixed part **21**.

[0064] (4) The second holding member **30** is fixed to the first holding member **20** by engagement between the engaging part **32** and the engaged part **23**. With this configuration, the second holding member **30** can be made attachable to and detachable from the first holding member **20** by the

engaging part **32** and the engaged part **23**.

[0065] (5) The engaged part **23** has the insertion hole **26**. The engaging part **32** has the pillar portion **35** that is inserted into the insertion hole **26**, and the protruding portion **36** that protrudes from the pillar portion **35** and is locked with the engaged part **23** in the counter-insertion direction. With this configuration, the engaging part **32** can be made attachable to and detachable from the engaged part **23**.

[0066] (6) There are a plurality of engaging parts **32** and a plurality of insertion holes **26**. With this configuration, the second holding member **30** can be firmly fixed to the first holding member **20** by the plurality of sets of engaging parts **32** and insertion holes **26**.

[0067] (7) The second holding member **30** has the engaging part **32**. The first holding member **20** has the engaged part **23** with which the engaging part **32** engages. Since the engaged part **23** provided in the first holding member **20** has the insertion hole **26**, it is possible to avoid the engaged part **23** from interfering with surrounding components when the first holding member **20** is used without the second holding member **30** attached.

[0068] (8) The protruding portion **36** has a plate-like shape that is elastically deformable in the direction along the insertion direction D with respect to the insertion hole **26**. With this configuration, when the engaging part **32** is inserted into or pulled out of the insertion hole **26**, the protruding portion **36** hits the wall portion **27** and elastically deforms. This makes it possible to reduce the force required to insert or pull the engaging part **32** into or out of the insertion hole **26**. As a result, the engaging part **32** including the protruding portion **36** can be suitably attached to and detached from the engaged part **23**.

[0069] (9) The protruding portion **36** is integrally formed with the pillar portion **35**. This configuration makes it possible to improve the locking strength of the protruding portion **36** to the engaged part **23**.

[0070] (10) One engaging part **32** is provided with a plurality of protruding portions **36**. This makes it possible to suppress the engaging part **32** from falling off from the engaged part **23**.

[0071] (11) The first electric wire holding part **22** has the band portion **24** that surrounds the outer periphery of the first electric wire member **11** and the lock portion **25** by which the band portion **24** is locked. In the first holding member **20**, the engaged part **23** is provided between the lock portion **25** and the fixed part **21**. With this configuration, the engaging part **32** can be preferably configured in the first holding member **20**.

[0072] The present embodiment can be modified as follows. The present embodiment and the following modified examples can be combined with each other to the extent that there is no technical contradiction. [0073] In the engaging part **32** of the above embodiment, the protruding portion **36** can be changed to a protruding portion **41** shown in FIG. 4 or a protruding portion **42** shown in FIG. 5, for example. The protruding portions **41** and **42** have a spherical or ellipsoidal shape. That is, the surfaces of the protruding portions **41** and **42** are spherical. With this configuration, the engaging part **32** including the protruding portion **41** or the protruding portion **42** can be suitably attached to and detached from the engaged part **23**. As shown in FIG. 4, the protruding portion **41** is integrally formed with the pillar portion **35**. This improves the locking strength of the protruding portion **41** to the engaged part **23**.

[0074] As shown in FIG. 5, the protruding portion **42** is configured as a separate body from the pillar portion **35**. More specifically, in the engaging part **32**, only the pillar portion **35** is formed integrally with the base portion **37**. The engaging part **32** shown in FIG. 5 is formed by attaching the protruding portion **42**, which is formed separately from the pillar portion **35**, to the pillar portion **35**. This configuration improves the formability of the engaging part **32**. [0075] In the engaged part **23**, the wall portion **27** may be a wall perpendicular to the Z axis, and the insertion hole **26** may pass through the wall portion **27** along the Z axis, for example. The wall portion **27** may be a wall perpendicular to the X axis, and the insertion hole **26** may pass through the wall portion **27** along the X axis, for example. The wall portion **27** may be a wall inclined with respect

to the Z axis, and the insertion hole **26** may be opened in an oblique direction with respect to the Z axis, for example. [0076] The number of the insertion holes **26** is not limited to that in the above embodiment and may be one or three or more. [0077] The number of the engaging parts **32** is not limited to that in the above embodiment and may be one or three or more. [0078] The number of the protruding portions **36** provided in one engaging part **32** is not limited to that in the above embodiment and may be one or three or more. [0079] The first holding member **20** may have the engaging part **32**, and the second holding member **30** may have the engaged part **23**. [0080] The configurations of the first electric wire holding part **22** and the second electric wire holding part **31** are not limited to those in the above embodiment, and may be changed as appropriate depending on the configuration. The configuration of the fixed part **21** is not limited to that in the above embodiment, and may be changed as appropriate depending on the configuration. [0081] The first holding member **20** can be changed to a binding member **51** shown in FIG. **6**. The binding member **51** is a member that binds an electric wire bundle **11a** of the first electric wire member **11**. The binding member **51** is an adhesive tape that is wound around the outer periphery of the electric wire bundle **11a**, for example. The material of the binding member **51** may be a resin material such as polyvinyl chloride (PVC), polypropylene (PP), or polyethylene terephthalate (PET), for example. [0082] The pillar portion **35** of the second holding member **30** passes through a through hole **52** formed in the binding member **51**. The through hole **52** is formed by breaking through the binding member **51** with the pillar portion **35**, for example. After passing through the through hole **52**, the protruding portion **36** of the engaging part **32** elastically returns to its original shape. Accordingly, the protruding portion **36** is locked on the inner peripheral surface of the binding member **51**. The second holding member **30** is held by the binding member **51** as the protruding portion **36** is locked to the binding member **51**. This configuration makes it possible to attach and detach the second holding member **30** to and from the binding member **51** that binds the electric wire bundle **11a**. [0083] The fixed part **21** in the illustrated embodiment can include the flange portion **21b** and a leg or plug configured to be inserted into the attachment hole Pa of the panel P. The flange portion **21b** can include a first flange surface configured to directly contact the attachment surface of the panel P and a second flange surface opposite to the first flange surface. The leg or plug may extend from the first flange surface of the flange portion **21b**. The leg or plug may include the projecting piece **21a** in the illustrated embodiment. [0084] The engaged part **23**, the lock portion **25**, the insertion hole **26**, and the wall portion **27** in the illustrated embodiment may be configured as a part of a first base block that protrudes from the second flange surface of the flange portion **21b**. The first base block may include a seat surface that supports the first electric wire member **11**. The first base block may cooperate with the band portion **24** in the illustrated embodiment to hold the first electric wire member **11** and position the first electric wire member **11** with respect to the panel P. [0085] The lock portion **34** and the base portion **37** in the illustrated embodiment may be configured as a part of a second base block that is attachable to and detachable from the first base block. The second base block may include a seat surface that supports the second electric wire member **12**. The second base block may cooperate with the band portion **33** in the illustrated embodiment to hold the second electric wire member **12** and position the second electric wire member **12** with respect to the panel P and the first electric wire member **11**. [0086] The first base block may support the second base block in a retrofittable manner with a friction engagement structure in between, which may be a recess and projection engagement structure, for example. In one example, the friction engagement structure may be configured such that the second base block retrofitted to the first base block is removable from the first base block. In one example, the first base block may support the second base block in a reversibly removable manner. [0087] Some implementation examples of the present disclosure are directed to a wire harness fixing device. The wire harness fixing device can be used together with an attachment target that has an attachment hole that may be a through hole, and the attachment target may be a part of a vehicle. The wire harness fixing device may include: a flange portion that has a first flange surface configured to

directly contact an attachment surface of the attachment target and a second flange surface that is opposite to the first flange surface; a plug that extends from the first flange surface of the flange portion; a first base block that extends from the second flange surface of the flange; and a first electric wire holding band that cooperates with the first base block to hold a first electric wire member and position the first electric wire member with respect to the attachment target. The first base block may support a second base block for holding a second electric wire member different from the first electric wire member in a retrofittable manner. The first base block may support the second base block with a friction engagement structure that may be a recess and projection engagement structure in between, for example. [0088] The embodiments and modifications disclosed herein should be considered to be illustrative and the present invention is not limited to these examples. The scope of the present invention is indicated by the scope of claims, and is intended to include all modifications within a meaning and scope equivalent to the scope of claims.

LIST OF REFERENCE NUMERALS

[0089] **10** Wire harness [0090] **11** First electric wire member [0091] **11a** Electric wire bundle [0092] **12** Second electric wire member [0093] **13** Wire harness fixing device [0094] **20** First holding member [0095] **21** Fixed part [0096] **21a** Projecting piece [0097] **21b** Flange portion [0098] **22** First electric wire holding part (electric wire holding part) [0099] **23** Engaged part [0100] **24** Band portion [0101] **25** Lock portion [0102] **26** Insertion hole [0103] **27** Wall portion [0104] **30** Second holding member [0105] **31** Second electric wire holding part [0106] **32** Engaging part [0107] **33** Band portion [0108] **34** Lock portion [0109] **35** Pillar portion [0110] **36** Protruding portion [0111] **37** Base portion [0112] **41,42** Protruding portion [0113] **51** Binding member (first holding member, adhesive tape) [0114] **52** Through hole [0115] **D** Insertion direction [0116] **P** Vehicle body panel (attachment target) [0117] **Pa** Attachment hole

Claims

1. A wire harness fixing device comprising: a first holding member that holds a first electric wire member; and a second holding member that holds a second electric wire member, wherein the second holding member is attachable to and detachable from the first holding member.
2. The wire harness fixing device according to claim 1, wherein the first holding member has a fixed part that is fixed to an attachment target and an electric wire holding part that holds the first electric wire member.
3. The wire harness fixing device according to claim 2, wherein the electric wire holding part is inseparable from the fixed part.
4. The wire harness fixing device according to claim 1, wherein one of the first holding member and the second holding member has an engaging part, the other of the first holding member and the second holding member has an engaged part with which the engaging part engages, and the second holding member is fixed to the first holding member by engagement between the engaging part and the engaged part.
5. The wire harness fixing device according to claim 4, wherein the engaged part has an insertion hole, and the engaging part has a pillar portion that is inserted into the insertion hole and a protruding portion that protrudes from the pillar portion and is locked to the engaged part in a counter-insertion direction.
6. The wire harness fixing device according to claim 5, wherein a plurality of the engaging parts and a plurality of the insertion holes are provided.
7. The wire harness fixing device according to claim 4, wherein the engaged part has an insertion hole into which the engaging part is inserted, and the engaged part is provided in the first holding member.
8. The wire harness fixing device according to claim 5, wherein the protruding portion has a plate-like shape that is elastically deformable in a direction along the insertion direction with respect to

the insertion hole.

9. The wire harness fixing device according to claim 5, wherein the protruding portion forms a spherical or ellipsoidal shape.

10. The wire harness fixing device according to claim 5, wherein the protruding portion is integrally formed with the pillar portion.

11. The wire harness fixing device according to claim 5, wherein the protruding portion is configured separately from the pillar portion.

12. The wire harness fixing device according to claim 4, wherein the first holding member has a fixed part that is fixed to an attachment target and an electric wire holding part that holds the first electric wire member, the electric wire holding part has a band portion that surrounds an outer periphery of the first electric wire member and a lock portion by which the band portion is locked, and in the first holding member, the engaging part or the engaged part is provided between the lock portion and the fixed part.

13. The wire harness fixing device according to claim 1, wherein the first holding member is a binding member that binds an electric wire bundle of the first electric wire member.

14. The wire harness fixing device according to claim 13, wherein the binding member is an adhesive tape that is wound around an outer periphery of the electric wire bundle.

15. A wire harness comprising: the wire harness fixing device according to claim 1; a first electric wire member that is held by the first holding member of the wire harness fixing device; and a second electric wire member that is held by the second holding member of the wire harness fixing device.
